"Calculus 3"

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 27

THE END

Any Reminders? Any Questions?

- Teaching evaluations are open!

- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8 ~~16.9~~

- Review for Exam 3 during class on Thursday!
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

ALVARO: Start the recording!

"Calculus 3"

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Review of Vector Calculus

Questions?

1. Vector Functions and Curves

P1. Find the unit tangent vector $\vec{T}(t)$ for $\vec{r}(t) = \langle t, \frac{2}{3}t^{3/2}, t^2 \rangle$ at $t = 1$.

P2. Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \le t \le \ln(2)$.

P3. Consider the curve C formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1 - y$.

(a) Find a parametrization $\vec{r}(t)$ for C.

(b) Find $\vec{r}'(t)$ and the line integral $\oint_C \langle -y, x, z \rangle \cdot d\vec{r}$.

2. Conservative Fields and Green's Theorem

P4. Use Green's Theorem to evaluate $\oint_C (xy + e^{x^2})\,dx + (x^2 + \sin(y^2))\,dy$ where C is the boundary of the rectangle with vertices $(0, 0)$, $(3, 0)$, $(3, 2)$, and $(0, 2)$.

P5. Show that $\vec{F} = \langle y\cos(xy), x\cos(xy) + 2y\rangle$ is conservative and find a potential function f. Use it to evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, \pi)$.

3. Parametric Surfaces and Surface Area

P6. Let S be the part of the plane $3x + 2y + z = 6$ in the first octant.

(a) Find a parametrization $\vec{r}(x, y)$ for S.

(b) Express the domain D as a **Type I (Type-x)** domain.

(c) Compute $\vec{r}_x \times \vec{r}_y$ and find the surface area of S using integration.

P7. Find the area of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 4$.

4. Flux and Vector Integrals

P8. Let S be the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

(a) Write a parametrization $\vec{r}(r, \theta)$ for S using **polar coordinates**.

(b) Find the normal vector $\vec{r}_r \times \vec{r}_\theta$.

(c) Evaluate the flux of $\vec{F} = \langle x, y, z^4 \rangle$ through S with upward orientation.

5. Stokes' and Divergence Theorems

P9. Let $\vec{F} = \langle z^2, x^2, y^2 \rangle$. Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where C is the triangle with vertices $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, oriented counterclockwise when viewed from above.

P1. Find the unit tangent vector $\vec{T}(t)$ for $\vec{r}(t) = \langle t, \frac{2}{3}t^{3/2}, t^2 \rangle$ at $t = 1$.

P2. Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \leq t \leq \ln(2)$.

P3. Consider the curve C formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1 - y$.

(a) Find a parametrization $\vec{r}(t)$ for C.

(b) Find $\vec{r}'(t)$ and the line integral $\oint_C \langle -y, x, z\rangle \cdot d\vec{r}$.

$$\int_0^{2\pi} F(r(t)) \cdot r'(t)\, dt$$

$$\begin{cases} x^2+y^2=1 \\ z=1-y \end{cases}$$

$0 \le t \le 2\pi$

$r(t) = (\cos t, \sin t, 1-\sin t)$

$r'(t) = (-\sin t, \cos t, -\cos t)$

$F(r(t)) = (-\sin t, \cos t, 1-\sin t)$

$F(r(t)) \cdot r'(t) = \sin^2 t + \cos^2 t - (1-\sin t)\cos t$

$= 1 - \cos t + \sin t \cos t$

$$\oint F\,dr = \int_0^{2\pi} 1 - \cos t + \sin t \cos t \, dt$$

$$\ldots$$

$$= \boxed{2\pi}$$

$$F = (-y, x, z)$$

$$\operatorname{Curl}(F) = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ -y & x & z \end{vmatrix}$$

$$= 0i - 0j + 2k$$

$$= \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

P4. Use Green's Theorem to evaluate $\oint_C (xy + e^{x^2})\,dx + (x^2 + \sin(y^2))\,dy$ where C is the boundary of the rectangle with vertices $(0,0)$, $(3,0)$, $(3,2)$, and $(0,2)$.

P5. Show that $\vec{F} = \langle y\cos(xy), x\cos(xy) + 2y\rangle$ is conservative and find a potential function f. Use it to evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0)$ to $(1,\pi)$.

1) $F = (P, Q) = (P, Q, 0)$

$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 \iff F$ conservative $\iff \text{curl}(F) = 0$

$\frac{\partial Q}{\partial x} = \cos(xy) - xy\sin(xy)$

$\shortparallel$

$\frac{\partial P}{\partial y} = \cos(xy) - yx\sin(xy)$

cons!

2) f, $\nabla f = F$, $\frac{\partial f}{\partial x} = P$, $\frac{\partial f}{\partial y} = Q$

$\int y\cos(xy)\,dx = \sin(xy) + C(y)$

$u = xy$

$(du = y\,dy)$

$\frac{\partial}{\partial y}$: $x\cos(xy) + C'(y) = x\cos(xy) + 2y$

$\boxed{f = \sin(xy) + y^2}$

POTENTIAL FN.

$$\boxed{\int_C \nabla f \cdot dr = f(B) - f(A) \quad \text{C goes from A to B}}$$

Fund. Thm. of Line Integrals!

$$\int_C F\, dr = \int_C \nabla(\sin(xy) + y^2) \cdot dr$$

$(0,0) \to (1,\pi)$

$$= f(1,\pi) - f(0,0)$$

$$= \sin(\pi) + \pi^2 - \sin(0) - 0^2 = \boxed{\pi^2}$$

P6. Let S be the part of the plane $3x + 2y + z = 6$ in the first octant.

(a) Find a parametrization $\vec{r}(x, y)$ for S.

(b) Express the domain D as a **Type I (Type-x)** domain.

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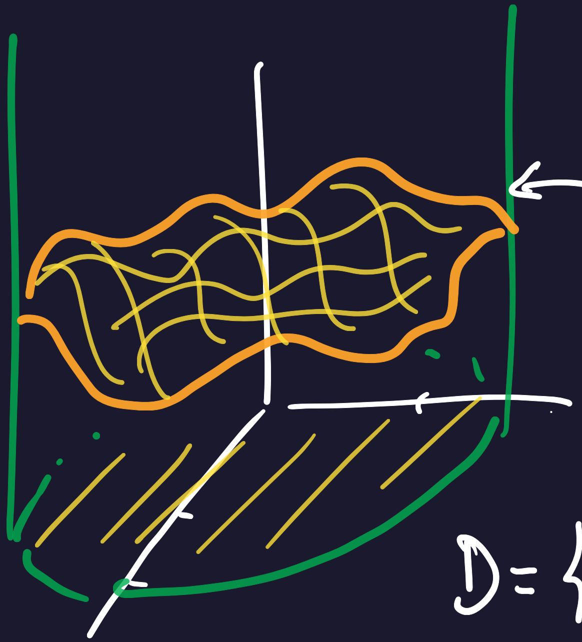


$z = f(x,y) = xy$

$S \rightsquigarrow r(u,v)$

$(u,v) \in D$

$D = \{x^2+y^2 \le 2\}$

$r(u,v) = (u, v, uv)$

$r_u = (1, 0, v)$

$r_v = (0, 1, u)$

$r_u \times r_v = \begin{vmatrix} i & j & k \\ 1 & 0 & v \\ 0 & 1 & u \end{vmatrix} = (-v, -u, 1)$

$|r_u \times r_v| = \sqrt{v^2 + u^2 + 1}$

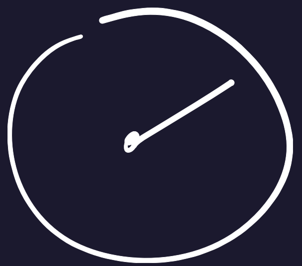
$$\iint_S 1 \, dS = \iint_D \|r_u \times r_v\| \, dA$$

$$= \iint_D \sqrt{1 + f_x^2 + f_y^2} \, dA$$

$f = xy, \quad f_x = y, \quad f_y = x$

$$= \iint_D \sqrt{1 + x^2 + y^2} \, dA$$

$$D = \{ x^2+y^2 \le 4 \}$$



$$D = \{ (r,\theta) : 0 \le r \le 2,\ 0 \le \theta \le 2\pi \}$$

$$\iint_D \sqrt{1+x^2+y^2}\, dA$$

$$\int_0^{2\pi} \int_0^2 \sqrt{1+r^2} \cdot \overbrace{r\, dr\, d\theta}^{dA}$$

$$= \underbrace{\int_0^{2\pi} d\theta}_{2\pi} \int_0^2 r\sqrt{1+r^2}\, dr$$

$$= 2\pi \int_0^2 r\sqrt{1+r^2}\, dr = 2\pi \, \frac{1}{2} \int_1^5 \sqrt{u}\, du = \dots = \boxed{\frac{2\pi}{3}\left(5\sqrt{5}-1\right)}$$

$u = 1+r^2$

$du = 2r\, dr$

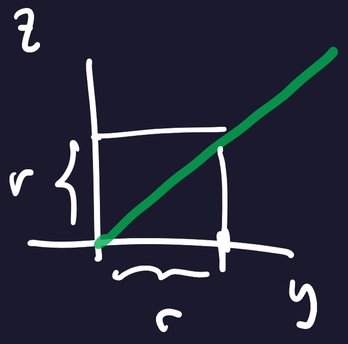
P8. Let S be the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

(a) Write a parametrization $\vec{r}(r, \theta)$ for S using **polar coordinates**.

(b) Find the normal vector $\vec{r}_r \times \vec{r}_\theta$.

(c) Evaluate the flux of $\vec{F} = \langle x, y, z^4 \rangle$ through S with upward orientation.

(a) $r(r, \theta) = (r\cos\theta, r\sin\theta, r)$



$D = \{(r, \theta) : 0 \le r \le 2, 0 \le \theta \le 2\pi\}$

$r_r = (\cos\theta, \sin\theta, 1)$

$r_\theta = (-r\sin\theta, r\cos\theta, 0)$

$$r_r \times r_\theta = \begin{vmatrix} i & j & k \\ \cos\theta & \sin\theta & 1 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (-r\cos\theta, -r\sin\theta, r)$$

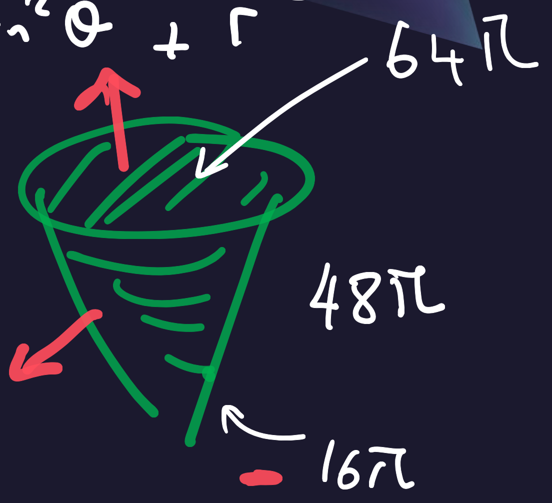
(c) $\iint_S F \cdot dS = \iint_D F(r(r,\theta)) \cdot (r_r \times r_\theta)\, dA$

$F = (x, y, z^4)$ $\quad r(r,\theta) = (r\cos\theta, r\sin\theta, r)$

$F(r(r,\theta)) = (r\cos\theta, r\sin\theta, r^4)$

$F(r(r,\theta)) \cdot (-r\cos\theta, -r\sin\theta, r) = -r^2\cos^2\theta - r^2\sin^2\theta + r^5$

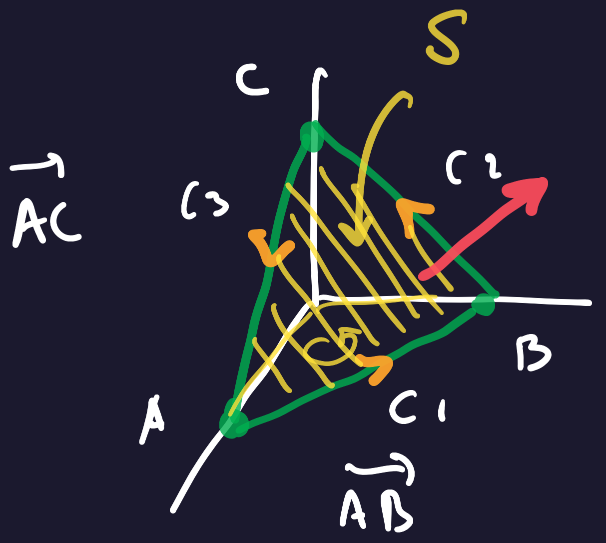
$= -r^2 + r^5$



$$\iint_S F \cdot dS = \int_0^{2\pi} \int_0^2 (-r^2 + r^5)\, dr\, d\theta$$

$$= \dots = \boxed{16\pi}$$

P9. Let $\vec{F} = \langle z^2, x^2, y^2 \rangle$. Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where C is the triangle with vertices $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, oriented counterclockwise when viewed from above.



$$\oint_C F\,dr = \int_{C_1} F\,dr + \int_{C_2} F\,dr + \int_{C_3} F\,dr$$

$\parallel$ STOKES

$$\iint_S \mathrm{Curl}(F)\,dS$$

$$\mathrm{Curl}(F) = \nabla \times F = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ z^2 & x^2 & y^2 \end{vmatrix} = (2y, 2z, 2x)$$

Parametrize.

$\vec{AC} = (-1, 0, 1)$

$\vec{AB} = (-1, 1, 0)$

$$\begin{vmatrix} i & j & k \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = (-1, -1, -1)$$

$n = (1, 1, 1)$

$n = (1, 1, 1)$

$p = (1, 0, 0)$

$1 \cdot (x-1) + 1 \cdot (y-0) + 1(z-0) = 0$

$x - 1 + y + z = 0$

$\boxed{x + y + z = 1}$

$z = 1 - x - y$

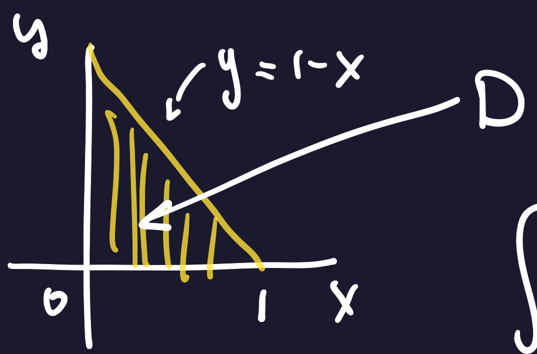
$r_u = (1, 0, -1)$

$r_v = (0, 1, -1)$

$(u, v, 1-u-v)$

$0 = 1 - x - y$

$y = 1 - x$



$D = \{(u,v) : 0 \leq x \leq 1,\ 0 \leq y \leq 1-x\}$

$$\iint_S \operatorname{Curl}(F)\, dS = \iint_D \nabla \times F(r(u,v)) \cdot n \, dA$$

$$\underbrace{\nabla \times F}_{(2y,\,2z,\,2x)}(r(u,v)) = (2v, 2(1-u-v), 2u) \cdot (1,1,1) = 2v + 2u + 2(1-u-v) = 2$$

$$\iint_D 2 \, dA = 2 \iint_D dA = 2\, A(D) = 2 \cdot \frac{1}{2} = \boxed{1}$$